



Product Information

Terostat-8590

One-component Adhesive/Sealant for
Direct Glazing of Vehicles, Railway Coaches, Ships and others

Basis: Polyurethane
Free of PVC and Solvents

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Characteristics

Terostat-8590 is a one component, moisture curing, pumpable adhesive/sealant based on polyurethane, which cures by reaction with moisture to a softelastic product. The skin formation and curing times are dependent on humidity and temperature, and the curing time also depends on joint depth. By increasing the temperature and moisture these times can be reduced; low temperature as well as low moisture retard the process.

The adhesive/sealant demonstrates the following characteristics:

- high shear strength, retained on ageing
- short stringing
- very high cure rate
- excellent adhesion on glass, ceramic coated glass (with Glass Primer Terostat-8511 or –8517) and on painted surfaces with (Paint Primer Terostat-8521 or Terostat-8522) or without primer dependent on paint type
- excellent adhesion on cut glazing adhesive/sealants, also on panes already pre-coated by the glass manufacturer with an adhesive/sealant based on PUR (using the Activator Terostat-8525 or Terostat-8526).

No active wipe prior to primer application with Terostat-8511 or –8517 is required, Cleaner Terostat-8550 is recommended for the screens. (When using other glass primers see the relevant Technical Data Sheets.)

Product Range

Terostat-8590 is available in a range of viscosities and containers:

Terostat-8590 N/M	low viscosity; delivered in cartridges, pails and drums
Terostat-8590 DK	medium viscosity; 310 ml cartridges for application by standard piston rod drive guns (air or hand operated).
Terostat-8590 UHV, 8590 V-25, 8590 KD-25	ultra-high viscosity; 310 ml cartridges (for application exclusively by special air operated guns), pails and drums.
Glazing Repair Set:	a complete repair set of materials for glazing comprising Terostat-8590 DK, cleaner, primer and equipment for screen removal and product application.

Application Areas

Terostat-8590 is used for the bonding of front, rear and side screens to the body of passenger cars, trucks, forklift and tractor cabins and special vehicles, busses, railwaycoaches, yachts, locomotives etc.

Terostat–8590 DK is mainly used for glazing repair, touch up on the production line and for small volume production.

Terostat–8590 UHV is a proved material because of its special characteristics (very high viscosity, short stringing, fast curing). Even when mounting heavy panes standing vertically, fixing devices can be eliminated in most cases.

Technical Data

Colour:	black
Odour:	barely perceptible
Consistency:	smooth, sag-resistant paste
Density:	1.20–1.28 g/cm ³
Solids:	100 %
Viscosity Data:	see graph 3
Curing mechanism:	moisture curing
Skin formation time:	ca. 25 mins (DIN 50014 standard climate: 23°C, 50 % rh)
Cure rate:	ca. 4–5 mm/24 h (DIN 50014 standard climate: 23°C, 50 % rh)
Shore–A–hardness (DIN 53505):	ca. 50
Tensile strength (DIN 53504):	ca. 7.5 MPa
Stress (DIN 53504):	ca. 2 MPa at 100 % elongation
Elongation at break (DIN 53504):	350–500 %
Shear strength:	ca. 2 MPa (after 24 h DIN 50014)
thickness 5mm, based on DIN 53283)	5–6 MPa (fully cured)
Tear resistance (DIN 53515):	ca. 6.5 N/mm
Volume change (DIN 52451):	< 1 %
Application temperature:	5°C to 35°C (50°C for UHV)
In service temperature range:	–40°C to 90°C
Short exposure (up to 1 h):	130°C

For variation of curing time, skin formation time, shear strength and viscosity properties with temperature and/or moisture see graphs below.

Additional data according to specifications of the automotive industry are available on request.

Application

Preliminary remark

Prior to application it is necessary to read the Safety Data Sheet for information about precautionary measures and safety recommendations. Also, for chemical products exempt from compulsory labelling, the relevant precautions should always be observed.

1. Application from Cartridges

Terostat–8590 DK can be directly applied from cartridges employing standard air or hand operated guns. Teroson recommends the use of

- Teroson Hand Pistol 340 (Art.No. 121.39 V)
- Teroson Staku Hand Pressure Pistol (Art.No. 167.65 Y)
- Telescope Pistol Power–Line (Art.No. 130.66 C)

Terostat–8590 UHV is heated to ca. 50°C to obtain a more suitable working viscosity. This heating can be provided in an air-circulation oven or a drying oven (duration: ca. 60 minutes) or to save time in the Teroson compact heater box (duration: ca. 15 minutes). The material is then applied immediately by means of a hand gun or compressed air gun (the latter with a piston rod drive). The equipment as above from the Teroson range may be used (except for the Teroson Hand Pistol 340) and the

- Teroson Heater Box (Art.No. 131.17 H)

2. Application from Pails and Drums

Terostat-8590 is applied from hobbicks and drums with high pressure pumps with follower plates, which must have the following characteristics:

- the entire equipment, including hoses and pipes, must be moisture tight
- the follower plate must be driven by a double pneumatic or hydraulic ram
- minimum pneumatic or hydraulic ratio: 48:1 for N/M 55:1 for UHV
- centering of the container.

The products are normally applied without any in line heating, however, a heating up to ca. 50°C for pipes and hoses is recommended for Terostat-8590 UHV.

Application is carried out either manually or by an automated application system (CNC or robot), often using heads especially designed to give particular bead characteristics. Teroson offers a head design capability and can recommend suitable suppliers of manual and automatic equipment.

A leaflet on handling Terostat-8590 from pails and drums is available from Teroson. This leaflet also includes detailed information about the steps in the direct glazing process.

3. Pretreatment of Surfaces

3.1 Cleaning

The surfaces to be bonded must be dry, free of oil, dust, grease and other contaminants.

Glass or ceramic coatings are cleaned with Cleaner Terostat-8550. Fresh painted surfaces normally do not require cleaning but the use of Cleaner Terostat-8550 is recommended when using stored or obviously contaminated pieces.

3.2 Priming

Surface priming is carried out by a thin (ca. 0.05 mm wet) and even coating of Glass Primer Terostat-8511 or -8517 on glass and ceramics and of Paint Primer Terostat-8521 or Terostat-8522 on paint.

It is recommended to carry out pre-production trials on each (ceramic coated) glass/paint combination. In many cases, particularly on 1-comp. alkyd or 2-comp. acrylic painted repair panels, it is not necessary to use the Paint Primer. Test of adhesion-properties are necessary.

Please contact Teroson for further information if required.

3.3 Repair

When a screen has been removed for repair a bead of cut and cured adhesive/sealant should be left on the body flange – do not use a primer on these surfaces. The cut surface provides an ideal substrate for application of fresh Terostat-8590 DK. Make sure not to contaminate the cut surface.

The instructions included in the Teroson Glazing Repair Set contain a detailed description of the repair operation.

3.4 Pre-coated screens

Screens for the repair job which have been pre-coated by the glass manufacturer with a PUR based adhesive/sealant must be treated with Activator Terostat-8525 or Terostat-8526 to ensure trouble-free adhesion of Terostat-8590 on to the pre-coated layer. The Activator is applied by felt applicator in a thin layer and allowed to air dry for 10–20 minutes. Terostat-8590 can then be applied in the normal manner but care must be exercised to take into account the 2 mm thickness of the pre-coating.

Screens pre-coated with PUR based adhesive/sealant are used for instance on many types of VW/Audi vehicles.

Test Certificate

Terostat-8590 DK is tested by the German TÜV. The respective test certificate of the TÜV-Südwest is available::



2 hours after glazing a vehicle without airbag can be driven again.

Storage

Frost-sensitive	no
Recommended storage temp.	10°C to 25°C
Shelf-life	12 months for –8590in cartridges 6 months in hobboks and drums (in original packaging each)

Packaging

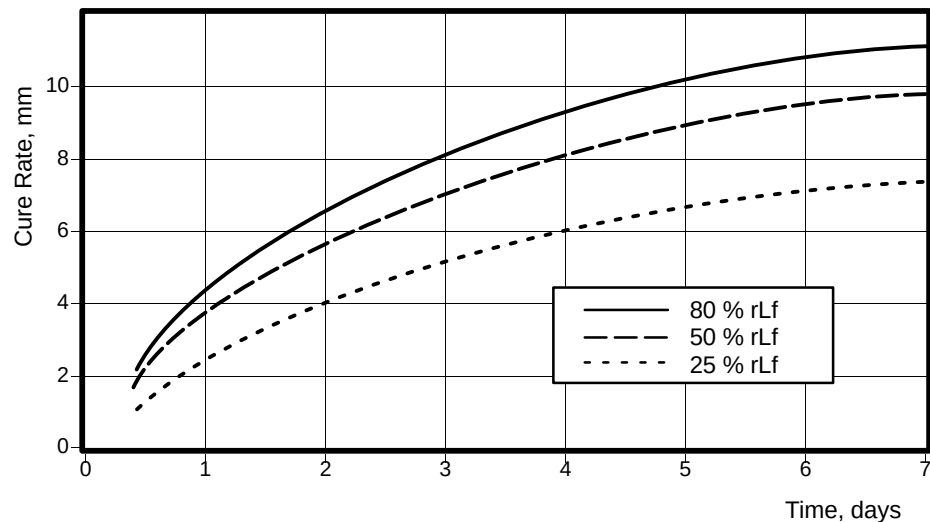
Cartridge	310 ml
Hobbock	30 kg ($\varnothing_i = 571,5$ mm, h = 880mm)
Drum	200 l ($\varnothing_i = 571,5$ mm, h = 880mm)
Direct Glazing Repair Set	

Hazard Indications/ Safety Recommendations/ Transport Regulations

see Safety Data Sheet

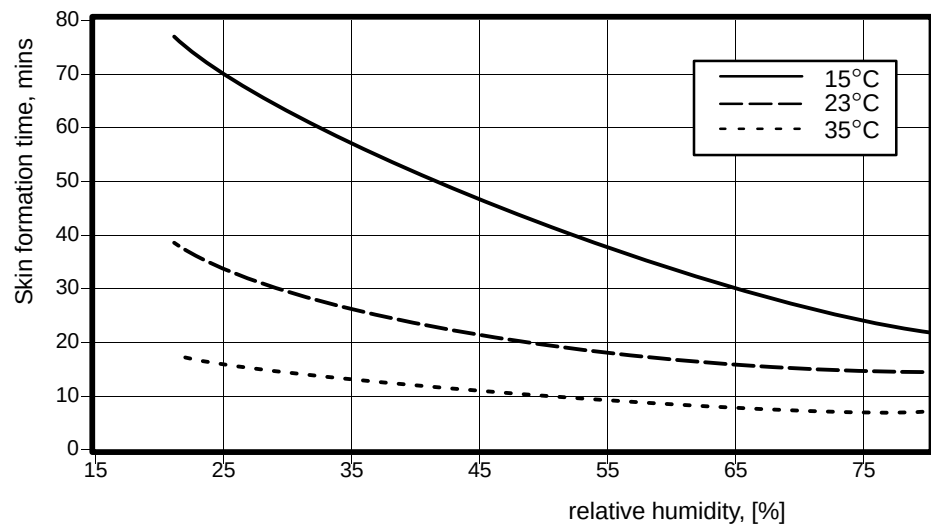
Graph 1

Cure rate of Terostat-8590 at 23°C and various Humidities



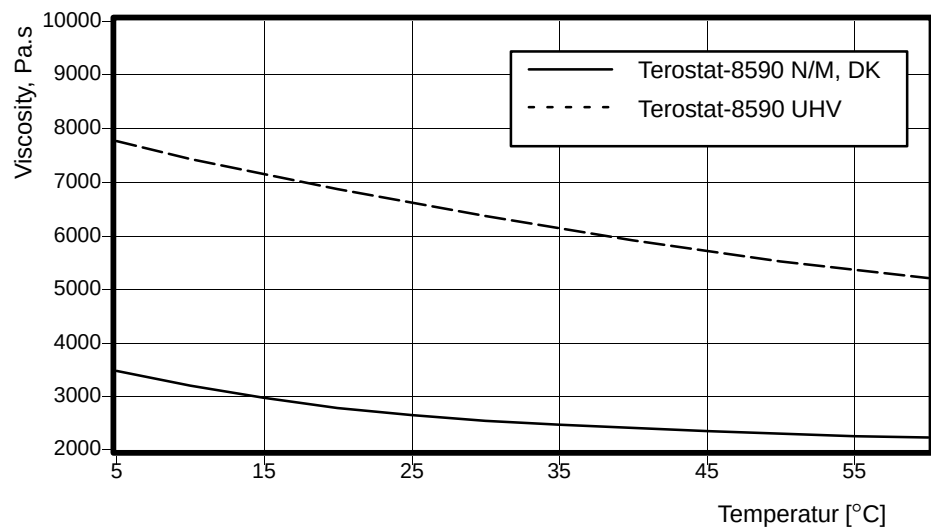
Graph 2

Skin formation time of Terostat-8590 as a function of the relative humidity at 3 temperatures



Graph 3

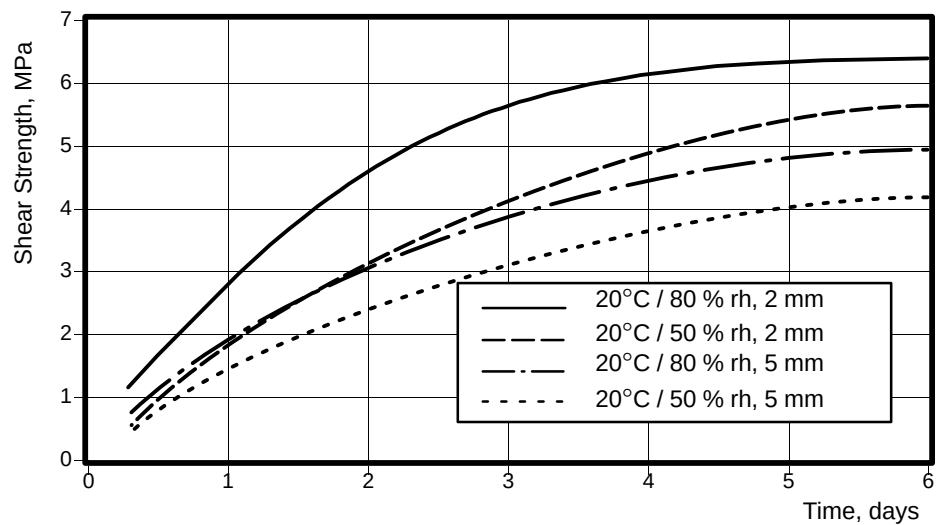
**Terostat-8590 N/M, DK, UHV
Relationship between viscosity and temperature**



Equipment: Viscolab MC 10; System Z-5 DIN; rate of shear 1/s
 Shear programme: 2 min 0 bis 1/s; 1 min bei 1/s; 2 min 1 bis 0/s
 Measured value: 1. Value on shear reduction curve at 1/s

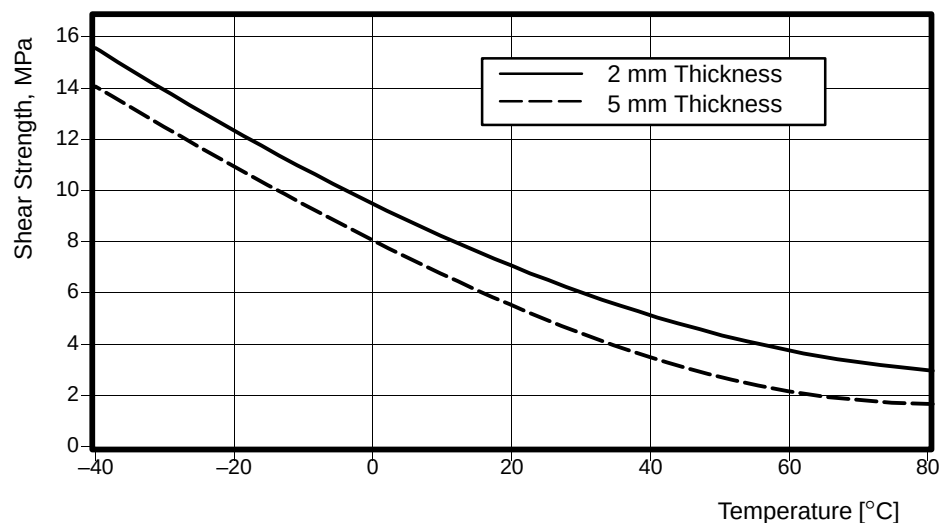
Graph 4

Shear strength of Terostat-8590 as a function of time at 2 humidities and 2 thicknesses



Graph 5

Shear strength of Terostat-8590 as a function of temperature at 2 thicknesses



Important

The above data, particularly the recommendations for application and use of our products are based on our knowledge and experience. Due to different materials and conditions of application which are beyond our knowledge and control we recommend strongly to carry out sufficient tests in order to ensure that our products are suitable for the intended processes and applications. Except for wilful acts any liability based on such recommendations or any oral advice is hereby expressly excluded.

This Technical Data Sheet supersedes all previous editions.

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